HDOC Day-48

Aim

To write a C program that finds the pivot integer between 1 and a given positive integer n using a function named **pivot_integer()** and pointers.

Problem Statement

Read a positive integer n and pass its address to the function **pivot_integer()**. Find an integer x between 1 and n such that the sum of integers from 1 to x is equal to the sum of integers from x to n. Display the pivot integer if it exists; otherwise display -1.

Algorithm

1. Start the program.
2. Read a positive integer n.
3. Pass the address of n to pivot_integer().
4. For every x from 1 to n, calculate the sum from 1 to x.
5. Calculate the sum from x to n.
6. If both sums are equal, display x and stop searching.
7. If no such x is found, display -1.
8. Stop the program.

C Program

```
#include <stdio.h>

void pivot_integer(int *n)
{
    int x, i;
    int leftSum, rightSum;
    int found = 0;

    for(x = 1; x <= *n; x++)
    {
        leftSum = 0;
        rightSum = 0;

        for(i = 1; i <= x; i++)
        {
            leftSum = leftSum + i;
        }

        for(i = x; i <= *n; i++)
        {
            rightSum = rightSum + i;
        }

        if(leftSum == rightSum)
        {
            printf("Pivot integer: %d\n", x);
            found = 1;
            break;
        }
    }
```

```
    if(found == 0)
    {
        printf("-1\n");
    }
}

int main()
{
    int n;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if(n > 0)
    {
        pivot_integer(&n);
    }
    else
    {
        printf("Please enter a positive integer.\n");
    }

    return 0;
}
```

Test Case Table

Test Case	Input	Expected Output
1	n = 8	Pivot integer: 6
2	n = 1	Pivot integer: 1
3	n = 4	-1

Output Snippets

Test Case 1

```
Enter a positive integer: 8
Pivot integer: 6
```

Test Case 2

```
Enter a positive integer: 1
Pivot integer: 1
```

Test Case 3

```
Enter a positive integer: 4
-1
```

Result

The program successfully finds and displays the pivot integer using pointers. If no pivot integer exists, it displays -1.